

Discrete Mathematics

Final Exam

December 16, 2022 (7:00pm - 9:30pm)

- Problems are printed on both sides of the paper (total 200 pts).
- You can leave after 8:00pm.
- Questions 1–5: Short-answer problems (No proofs are required).

Question 1. (20 points)

- (a) How many ways are there to choose 12 cookies if there are five varieties of cookies?
- (b) How many onto functions are there from a set with six elements to a set with four elements?

Question 2. (20 points) Two identical bags contain balls. One of the bags has 6 red balls and 3 blue balls. The other bag has 5 red balls and 8 blue balls. A bag is chosen at random and two balls are drawn at random from this bag, without replacement.

- (a) What is the probability that both balls are red?
- (b) What is the probability that the second ball is red, given that the first ball is red?

Question 3. (20 points) Consider the following relations on the set of positive integers.

$$\begin{aligned}R_1 &= \{(x, y) \mid x + y > 10\}, \\R_2 &= \{(x, y) \mid y \text{ divides } x\}, \\R_3 &= \{(x, y) \mid \gcd(x, y) = 1\}, \\R_4 &= \{(x, y) \mid x \text{ and } y \text{ have the same prime divisors.}\}.\end{aligned}$$

- (a) Which of these relations are reflexive?
- (b) Which of these relations are symmetric?
- (c) Which of these relations are antisymmetric?
- (d) Which of these relations are transitive?

Question 4. (15 points) How many nonisomorphic trees are there with five vertices? Draw them.

Question 5. (20 points) Consider the recurrence relation $a_n = -3a_{n-1} + 18a_{n-2} + 3^n$.

- (a) Find the general solution to the associated homogeneous recurrence relation.
- (b) Find a particular solution to the given recurrence relation.

Question 6. (20 points) Show that the closure of a relation R with respect to a property $\mathbf{P}$, if it exists, is the intersection of all the relations with property $\mathbf{P}$ that contain R .

Question 7. (20 points) Show that in every simple graph there is a path from every vertex of odd degree to some other vertex of odd degree.

Question 8. (20 points) What is the variance of the number of fixed elements, that is, elements left in the same position, of a randomly selected permutation of n elements?

Question 9. (20 points) Let $G = (V, E)$ be a connected simple graph. The following is a pseudocode of breadth-first search.

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procedure BFS( $G$ : connected graph with vertices  $v_1, v_2, \dots, v_n$ )  
   $T :=$  tree consisting only of vertex  $v_1$   
   $L :=$  empty list  
  put  $v_1$  in the list  $L$  of unprocessed vertices  
  while  $L$  is not empty  
    remove the first vertex,  $v$ , from  $L$   
    for each neighbor  $w$  of  $v$   
      if  $w$  is not in  $L$  and not in  $T$  then  
        add  $w$  to the end of the list  $L$   
        add  $w$  and edge  $\{v, w\}$  to  $T$ 
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(a) Show that all edges of G connect vertices at the same level or at adjoining levels of the spanning tree obtained by BFS.

(b) Show that the complexity of BFS is $O(|E|)$.

Question 10. (25 points) Consider a variant of the scheduling problem:

- Suppose we have a group of n proposed talks, where talk j begins at time s_j , ends at time e_j , and will be attended by w_j students.
- There are two lecture halls to schedule talks. A talk cannot be presented in both lecture halls.
- In each lecture hall, no two talks can proceed at the same time. Once a talk starts, it continues until it ends.

We want a schedule that maximizes the total number of student attendees. That is, we wish to schedule two subsets of talks to maximize the sum of w_j over all scheduled talks. Devise a dynamic programming algorithm for this problem.